All Models — Winner-per-M Comparison

Every ranking approach built in this project, side by side

The models

Seven distinct ranking approaches were built over the course of this project. They differ on three axes: which *families* they consider, whether they apply a *hard safety gate* (reject unsafe designs outright) or a *soft score* (penalize but never exclude), and what *objective* they rank survivors by.

Model	**Families**	**Gate**	**Ranks by**
metric1 (hard-gated)	all 10	margin-aware (real threshold, not > 0)	packing_density = N_{actual}/hull_area
FINAL_WINNER	square, hex, rectangular only	margin-aware (same as metric1)	packing_density (identical objective, narrower family scope)
S_paper	all 10	none (soft only)	Shen's closed-form $S = a_{\text{norm}} + \mu_\eta + \mu_g +$ cov
S_paper_soft	square, hex only (blind_rate undefined elsewhere)	none (soft only)	same, cov replaced by Shen's closed-form $1 -$ blind_rate
eff. density $c_{\min}$=0	all 10 (stochastic seed-aware)	margin-aware, optical-only threshold	effective_density = $n_{\text{active_holes}}$/hull_area
eff. density $c_{\min}$=55	all 10 (stochastic seed-aware)	margin-aware, Shen's perceptual floor	effective_density (same, floored threshold)
Shen MC evaluator	all 10	none (soft only)	Shen's own Sec. 2.4 Monte Carlo score S (1000 trials/candidate)

Master table: winner per M, every model

M	metric1 / FINAL_WINNER	S_paper	S_paper_soft	eff. density $c_{\min}$=0	eff. density $c_{\min}$=55	Shen MC
0.0	square, N=9, pitch=2000	archimedean_spiral, N=50, c=1500	square, N=20, pitch=4000	hexagonal, N=7, pitch=2000,30°	ring_offset_conc., N=11, rs=1000	poisson_disk, N=20, sf=2.0
1.0	square, N=100, pitch=3000,45°	poisson_disk, N=20, sf=2.0	hexagonal, N=20, pitch=3000	ring_offset_conc., N=20, rs=4000	square, N=20, pitch=2000	jittered_lattice, N=100, pitch=3000
2.0	square, N=100, pitch=3000,45°	poisson_disk, N=20, sf=2.0	hexagonal, N=20, pitch=2000	ring_offset_conc., N=20, rs=4000	ring_offset_conc., N=20, rs=4000	archimedean_spiral, N=20, c=1000
3.0	square, N=100, pitch=3000,45°	poisson_disk, N=20, sf=2.0	hexagonal, N=20, pitch=2000	ring_offset_conc., N=20, rs=4000	ring_offset_conc., N=20, rs=4000	archimedean_spiral, N=20, c=1000
4.0	square, N=50, pitch=3000,45°	jittered_lattice, N=50, pitch=1500	hexagonal, N=20, pitch=1500	ring_offset_conc., N=20, rs=4000	ring_offset_conc., N=20, rs=4000	archimedean_spiral, N=20, c=1000
5.0	square, N=50, pitch=3000,45°	jittered_lattice, N=50, pitch=1500	hexagonal, N=20, pitch=1500	ring_offset_conc., N=20, rs=4000	ring_offset_conc., N=20, rs=4000	archimedean_spiral, N=20, c=1000
6.0	square, N=50, pitch=3000,45°	poisson_disk, N=20, sf=2.0	hexagonal, N=20, pitch=1500	ring_offset_conc., N=20, rs=4000	ring_offset_conc., N=20, rs=4000	archimedean_spiral, N=20, c=1000
8.0	square, N=50, pitch=3000,45°	poisson_disk, N=20, sf=2.0	hexagonal, N=20, pitch=1000	ring_offset_conc., N=20, rs=4000	ring_offset_conc., N=20, rs=4000	poisson_disk, N=20, sf=2.0
global	square, N=9, pitch=2000	archimedean_spiral, N=50, c=1500	square, N=20, pitch=4000	hexagonal, N=7, pitch=2000,30°	hexagonal, N=7, pitch=2000,30°	*n/a (per-M only)*

Table 2: Winner per M, all seven models. Highlighted : the only cell where the $c_{\min}$ floor actually changes eff. density's own winner (M=0.0 also changes family but stays deterministic). "sf"=spacing_factor, "rs"=ring_spacing_um, "c"=c_um. N is N_requested; every row here had N_actual=N_requested except metric1/FINAL_WINNER/eff. density's M=0/global rows (N_actual=7–11, overlap-rejected).

Family that wins, at a glance

Model	0.0	1–3	4–5	6	8	global
metric1 / FINAL_WINNER	square	square	square	square	square	square
S_paper	arch. spiral	poisson/poisson/poisson	jittered	poisson	poisson	arch. spiral
S_paper_soft	square	hexagonal	hexagonal	hexagonal	hexagonal	square
eff. density $c_{\min}$=0	hexagonal	r.o. concentric	r.o. concentric	r.o. concentric	r.o. concentric	hexagonal
eff. density $c_{\min}$=55	r.o. concentric	**square**/r.o.c./r.o.c.	r.o. concentric	r.o. concentric	r.o. concentric	hexagonal
Shen MC	poisson	jittered/arch./arch.	arch. spiral	arch. spiral	poisson	—

Differences and reasons

1. metric1 and FINAL_WINNER are identical

FINAL_WINNER restricts the candidate pool to square/hexagonal/rectangular only; metric1 searches all 10 families. They produce the *same* winner at every M because square already wins metric1's unrestricted search everywhere — the family restriction never actually bound. This confirms square's dominance under packing_density + margin-aware gating is not an artifact of excluding other families.

2. S_paper and S_paper_soft disagree with metric1 almost everywhere, and with each other

Both are *soft* scores — no candidate is ever excluded, only down-weighted. Investigated and confirmed empirically (not assumed): g, the input to S_paper's ghost term μ_g, has mean ≈ 419 across this dataset's actual (spacing, blur) scale — always $\gg 1$ — so $\mu_g = \text{sigmoid}(2(1-g))$ is numerically $< 10^{-10}$ for **100% of 8501 rows**. The ghost term contributes nothing, safe or unsafe alike. What is left driving S_paper's ranking is mostly μ_η (closeness of fill-factor to the paper's target $\eta = 0.15$) plus the coverage term — properties orthogonal to whether a design actually ghosts. S_paper_soft additionally swaps the coverage term for a fixed *closed-form* blind-rate constant per family ($1 - \pi/4 \approx 78.5\%$ open for square, $1 - \pi/(2\sqrt{3}) \approx 90.7\%$ for hex) — hexagonal's better constant, combined with its own μ_η advantage, is enough to flip most M slots to hexagonal even though this project's own independently-computed vignetting margin for that exact hexagonal candidate is a real, decisive failure (-7 to $-20\,\mu$m of margin, verified directly against `sweep_results_with_density.csv`) at every M it wins. **Neither S_paper score has any vignetting term at all** — this is not a rounding issue, it is a structural gap in the formula as specified.

3. The Shen Monte Carlo evaluator disagrees with S_paper's own closed form

Correlation(ghost_rate, g_{Sch}) across all 7708 per-M candidates $= -0.0485$ — computed directly, not assumed. ghost_rate is bimodal (52.6% of candidates at exactly 1.0, 10.9% at exactly 0.0); g_{Sch} is a smooth unbounded number with no such structure. Shen's own MC winner (archimedean_spiral, M=2–6) independently reads ghost_rate 0.47–0.91 under Shen's *own* more rigorous method — Shen's own published optimizer and Shen's own MC evaluator disagree about Shen's own winning design. Separately, metric1's square winner shows an elevated Shen ghost_rate (0.38–0.82 at M=1–8) despite a real $+\infty$ margin under our own protocol — explained entirely by Shen's much more permissive "any nonzero overlap admits a beam" admission rule versus this project's own 50%-overlap `active_frac_threshold`; not an actual safety failure in our winner.

4. Effective density abandons square and hexagonal-by-S_{paper} entirely

metric1's packing_density counts every generated hole, including ones that structurally never admit light — confirmed: mean(effective_density / packing_density) $= 0.175$ across the whole dataset, i.e. packing_density over-counted by $\approx 5.7\times$ on average. Once ranking is restricted to *only* holes that actually admit light, square's dense-but-mostly-dark lattices lose to `ring_offset_concentric` (M$\geq$2) and `hexagonal` (M=0/global) — sparser designs whose few holes are each doing real work. This is the correct outcome of fixing a metric that rewarded dark holes, not a defect.

5. $c_{\min} = 55$ changes exactly one winner, and only after a real bug was fixed

`poisson_disk` (seed 7) initially won M=1.0 once Shen's perceptual floor made its margins positive — but investigation found this used the single *luckiest* seed of eight, only 2 of which actually cleared both floored margins (worst-case ghost margin across seeds $= -0.138$, seed-to-seed std $= 0.220$, flagged `high_seed_variance` by this project's own established

$> 1.0 \times |\text{worst}|$ rule). Fixed by grouping stochastic candidates (`poisson_disk`, `jittered_lattice`) by (family, N, spacing, M) and requiring the *worst* seed — not the best — to clear both margins before the group is even eligible, then ranking eligible groups by mean effective density across seeds that also pass viability. Result: **poisson_disk loses**; every remaining stochastic group that is technically margin-eligible still fails viability across every one of its seeds. The M=1.0 winner reverts to a deterministic family (`square`, N=20, pitch=2000) — the only cell in the whole comparison where the $c_{\min}$ floor changes eff. density's own answer at all.

Summary

No two of these seven models agree on a winner at every M. The disagreements are not noise — each is traceable to one specific, confirmed mechanism: a saturated formula term (S_paper/S_paper_soft's μ_g), a missing safety dimension (no vignetting term in either Shen score), a different beam-admission threshold (Shen MC vs. this project's 50% rule), a metric that counted dark holes as density (packing_density vs. effective_density), or a single-lucky-seed selection bug (fixed, poisson_disk → square at M=1.0). This project's own hard-gated pipeline (metric1/FINAL_WINNER) and its coverage-corrected successor (effective density, seed-aware) are the two models built on real, independently verified margins with an actual safety floor — the other five are useful comparison points, not competing safety-verified answers.